PUSAN NATIONAL UNIVERSITY

Dept. of Aerospace Engineering

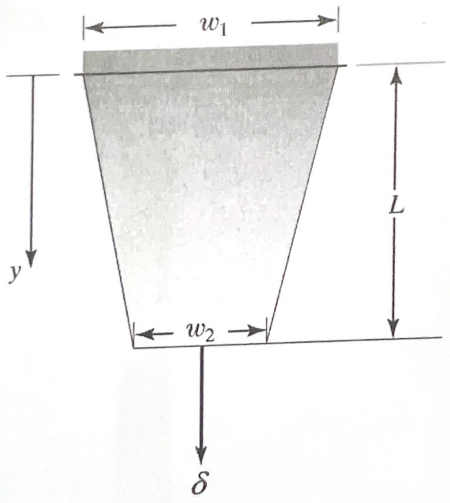
Computational Structural Analysis

Mid-exam. Oct. 27th, 2021

NOTICE: Write down the ANSYS commands on your answer sheet.

Problem 1

Consider a bar with variable cross section imposing a prescribed displacement δ in at $y = L$. Determine the deflection in term of δ at $y = 7.5$ in. The plate is 0.125 in thick and has a modulus of elasticity $E = 10.4 \times 10^6$ lb/in^2. Assume $w_1 = 2$ in, $w_2 = 1$ in, $L = 10$ in and use only **LINK180** as an element type to create the finite element model.

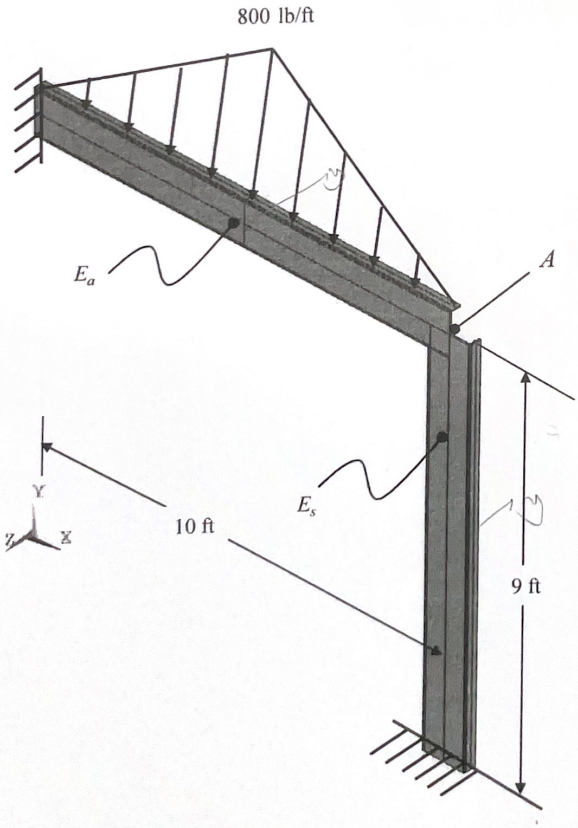


* 모든 문제에 R=287J/kg.K 과 γ=1.4를

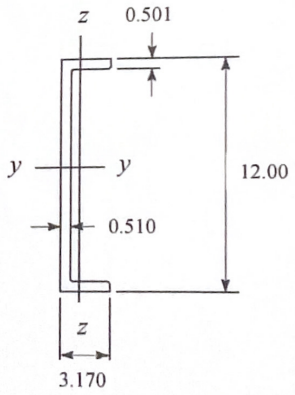
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Problem 2

Consider the overhang frame with the same cross section for the two members shown in below figure. The frame is made of two different materials steel and aluminum with $E_s = 30 \times 10^6$ lb/in^2 and $E_a = 10 \times 10^6$ lb/in^2, respectively. The frame is fixed as shown in the figure. Determine the deformation at the position *A* of the frame under the given gradually varying distributed load using an element type of **BEAM188**.



[units : in]



y, *z* ; element coordinate system